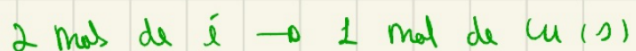
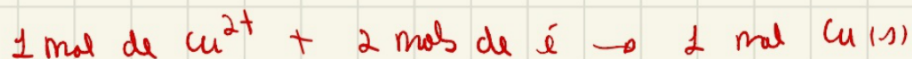
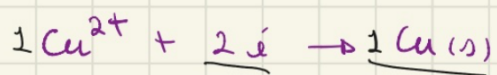


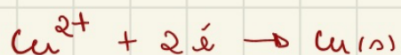
* ESTEQUIOMETRIA E ELETROLÍSE (LEI DE FARADAY)



10 mols de e^- $\rightarrow$ qual é a massa de cobre formada?

$$\text{Cu} = 63,5 \text{ g/mol}$$

$$63,5 \text{ g} \rightarrow 1 \text{ mol de Cu(s)}$$



$$x \rightarrow 5 \text{ mols de Cu(s)}$$

$$2 \text{ mols e}^- \rightarrow 1 \text{ mol Cu(s)} \quad 2x = 10$$

$$x = 5 \cdot 63,5$$

$$10 \text{ mols e}^- \rightarrow x$$

$$x = 5 \text{ mols de Cu(s)}$$

$$x = 317,5 \text{ g de Cu(s)}$$

* Como pode APARECER em QUESTÕES?

$$1F = 1 \text{ mol de } e^-$$

$$1 \text{ FARADAY} = 96500 \text{ C}$$

$$1 \text{ mol de } e^- = 96500 \text{ C}$$

EX: UMA SOLUÇÃO DE Cu^{2+} FOI RECARREGADA COM UMA CORRENTE ELÉTRICA DE 965 AMPERES POR UM TEMPO DE 10 segundos.

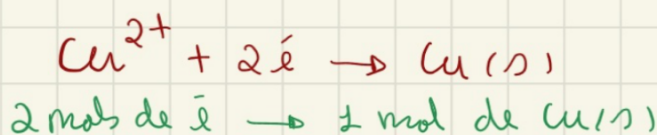
A PARTIR DISSO, QUAL É A MASSA DE Cu(s) FORMADA?

$$\text{Cu} = 63,5 \text{ g/mol}$$

$$1F = 96500 \text{ C}$$

$$Q = i \cdot t$$

↓ ↓ segundos
CARREGA (C) (A)



$$Q = 965 \cdot 10 \Rightarrow Q = 9650 \text{ C}$$

$$1F = 96500 \text{ C}$$

$$X = 9650 \text{ C}$$

$$X = \frac{1}{10}$$

$$X = 0,1 F$$

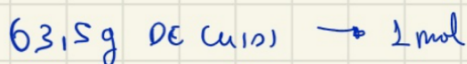
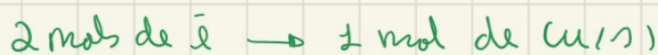
$$1F = 1 \text{ mol de } e^-$$

$$0,1 F$$

$$0,1 \text{ mol } e^-$$

$$X = \frac{9650}{96500}$$

$$X = \frac{965}{9650} \Rightarrow X = \frac{965}{965 \cdot 10}$$



$$Y = \frac{0,1}{2}$$

$$Y = 0,05 \text{ mols de Cu(s)}$$

$$X = 63,5 \cdot 0,05$$

$$X = 3,175 \text{ g de Cu(s)}$$